

RUPAL CHAUHAN

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SUMMARY

Bioinformatics researcher with a Ph.D. and 5+ years of experience in genomic data analysis, NGS pipelines, and structure-based drug design. Skilled in developing computational workflows, building biological databases, and collaborating with cross-functional teams at the FDA and pharmaceutical companies. Experienced in applying Python, R, and molecular modeling tools to accelerate biomarker and drug discovery.

PROFESSIONAL EXPERIENCE

Researcher II – Midnight Sun Technologies (FDA, Silver Spring, MD) | Nov 2022 – Jul 2023

- Revamped FDA's Site Selection Model (SSM) by integrating new datasets and analytical methods, improving selection accuracy and supporting regulatory decision-making.
- Authored scientific manuscripts, technical reports, and documented unmet data needs of stakeholders.
- Collaborated with cross-functional teams to identify gaps, propose solutions, and implement enhancements.

Sr. Technical Writer – Catalent Pharma Solutions, Manassas, VA | Mar 2022 – Nov 2022

- Authored and revised master batch records and technical documentation in compliance with GMP standards.
- Prepared material and product specifications, supporting implementation of new manufacturing tools.

LIMS Associate & Technical Writer – Granules Pharmaceutical Inc., Chantilly, VA | Oct 2021 – Mar 2022

- Managed clinical information in LIMS database, triaged specimens, and conducted internal audits.
- Streamlined documentation under GMP change control system, reducing errors and ensuring compliance.

EDUCATION

Ph.D. in Bioinformatics, Gujarat University, India (2017) Dissertation: In silico and molecular analysis of cereal crops; developed a database of fruit crop diseases using MySQL/PHP.

TECHNICAL SKILLS

Programming & Data Analysis: Python, R, Perl, C, C++, Java, SQL, ORACLE **Bioinformatics Tools:** BLAST, CLUSTAL, MEGA, MEME, AutoDock, i-TASSER, PyMOL **NGS & Omics:** RNA/DNA sequencing, proteomics, metabolomics pipelines **Databases & Web Development:** MySQL, PHP, HTML, CSS **Other:** LIMS, GMP documentation, MS Office

PUBLICATIONS

- Chauhan R. et al. (2018). Molecular characterization of L. donovani XPRT and evaluation of inhibitors. Biochim Biophys Acta - Proteins & Proteomics.

- Chauhan R. et al. (2015). Genetic diversity of cereal crops using RAPD markers. Journal of Bioinformatics & Proteomics Review.
- Chauhan R. et al. (2014). FCDD: A Database for Fruit Crops Diseases. Bioinformation.
- Chauhan R. et al. (2014). In Silico Analysis for Five Major Cereal Crops Phytocystatins. Interdisciplinary Sciences Computational Life Sciences.